

Q1. I have recently undertaken a project on streaming video over Bluetooth. Please guide me on the Bluetooth theory, programming and streaming.

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A1. Bluetooth is an exciting technology for mobile devices that serves the purpose of streaming video in ad hoc network environments. You have taken a project in an emerging field of technology, which provides good opportunity for research.

Bluetooth supports three kinds of connection schemes for forming ad hoc networks: point-to-point, piconet and scatternet. For video streaming over Bluetooth link, you will have to address three major technologies, viz. video compression, quality-of-service (QoS) control and intermediate protocols. Each of these areas is one of the basic components in building a complete architecture for streaming video over Bluetooth. The relationship among them is shown in Fig. 1.

Fig. 1 shows the functional components for video streaming over Bluetooth links. Moreover, the layers over which a component works are also indicated. The aim of video compression is to remove redundant information from a digitised video sequence. Raw data must be compressed before transmission to achieve efficiency. This is critical for wireless video streaming since the bandwidth of wireless links is limited to 732 kbps.

Upon the client's request, the media server retrieves compressed video and the QoS control module adapts the media bitstreams, or adjusts transmission parameters of the intermediate layer based on the current link status and QoS requirements. After the adaptation, compressed video streams are partitioned into packets of the chosen intermediate layer (L2CAP, HCI, IP), where packets are prepared and segmented. It then sends the segmented packets to the Bluetooth module for transmission.

On the receiving side, the Bluetooth module receives media packets from

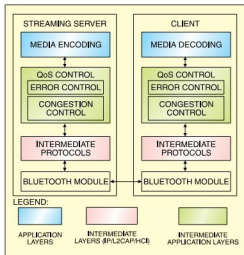


Fig. 1: Architecture for streaming over Bluetooth

the air, reassembles them in the intermediate protocols and sends them to the decoder for decompression.

As shown in Fig. 1, QoS control can be further categorised into congestion control and error control. Congestion control in Bluetooth is employed to prevent packet loss and reduce delay by regulating the transmission rate or reserving the bandwidth according to the changing link status and QoS requirements. Error control, on the other hand, improves the video quality in the presence of packet loss.

You may read an article titled 'Video Streaming Over Bluetooth: A Survey' by Wang Xiaohang for more information (<http://www.comp.nus.edu.sg/~cs5248/0304S1/surveys/wang-bluetooth.pdf>). Also there are a number of books on the subject. The subject is under research and you need to have thorough knowledge of the Bluetooth radio architecture—both hardware and software.

Q2. What is the difference between a normal speaker, piezobuzzer and horn speaker?

Deepak Sharan
Through e-mail

A2. A loudspeaker, or speaker system, is an electroacoustical transducer that converts an electrical signal into sound. The term loudspeaker can refer to individual transducers

(known as drivers), or to complete systems consisting of an enclosure incorporating one or more drivers and electrical filter components as shown in Fig. 2. Loudspeakers, just as with other electroacoustic transducers, are the most variable elements in an audio system and responsible for the greatest degree of audible differences between sound systems.

To adequately reproduce a wide range of frequencies, most loudspeaker systems require more than one driver, particularly for high sound pressure level or high accuracy. Individual drivers

are used to reproduce different frequency ranges. The drivers are named subwoofers (for very low frequencies),

woofers (for low frequencies), mid-range speakers (for middle frequencies), tweeters (for high frequencies) and sometimes su-



Fig. 2: Speaker system

per-tweeters (optimised for the highest audible frequencies).

The piezoelectric sound components operate on an innovative principle utilising the natural oscillation of piezoelectric ceramics. These are applied widely in alarms, speakers, telephone ringers, receivers, transmitters or buzzers. You have asked about the normal speakers, piezobuzzers and horn speakers. Therefore we are describing both the piezo speakers as well as piezobuzzers for your understanding.

Piezoelectric speakers. Piezoelectric speakers are frequently used as beepers in watches and other electronic devices, and sometimes as tweeters in less-expensive speaker systems, such